

Wireless CALL BELL

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Traditional doorbell systems often require extensive wiring, intricate assembly, and the expertise of a qualified electrician for installation. In contrast, wireless doorbells offer a simpler installation process compared to their wired counterparts. With a lack of complicated wiring, a wireless call bell can be easily installed by anyone within a household. This article demonstrates the process. Here we incorporate an encoder and decoder for the creation of a wireless call bell system. The prototypes for the author's transmitter and receiver are depicted in Figs 1 and 2 respectively.

Circuit and working

This wireless call bell utilises a 433MHz RF transmitter and a receiver in two separate modules. The transmitter facilitates the transmission of a 433MHz RF signal, while the receiver is responsible for receiving the signal.

Transmitter. The transmitter employs an encoder IC in conjunction with a switch, enabling the activation of the call bell. Fig. 3 illustrates the circuit diagram for the transmitting unit. The circuit features an encoder HT12E (IC1), a 433MHz RF transmitter (TX1), and other essential components. The encoder HT12E encodes the transmission signal, with its address data encoding set via the active low address pins (A0 through A7) of encoder IC1. The address pin of the encoder corresponds to that of decoder IC2 in the receiver unit. Mismatched addresses between the encoder and decoder prevent signal

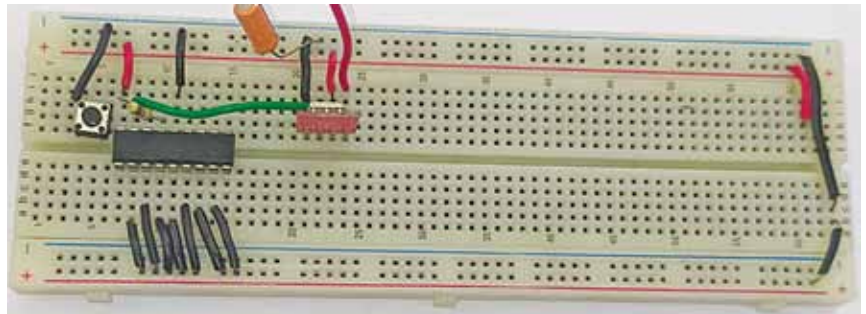


Fig. 1: Author's transmitter prototype

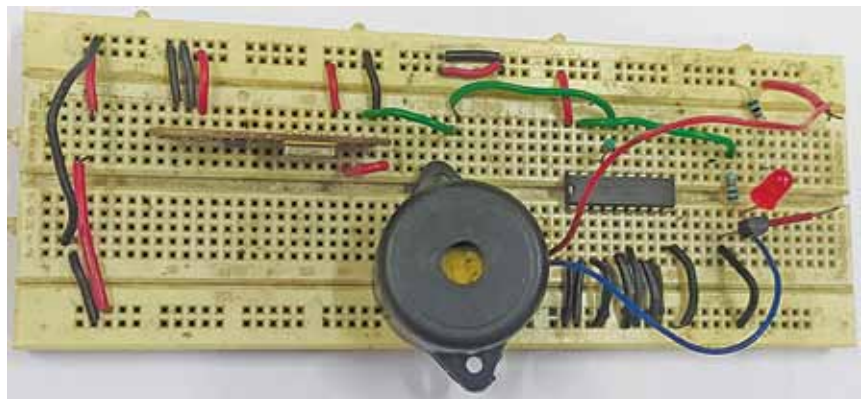


Fig. 2: Author's receiver prototype

reception. When switch S1 is momentarily pressed, the Dout (pin 17) emits a signal (data) through the RF transmitter (TX1). This encoded data is subsequently received by the receiver unit.

Receiver. The receiver employs a decoder IC along with a buzzer for signal reception sent by the transmitter. The circuit diagram of the receiver unit is depicted in Fig. 4. The circuit includes a 433MHz RF receiver (RX1), decoder HT12D (IC2), npn transis-

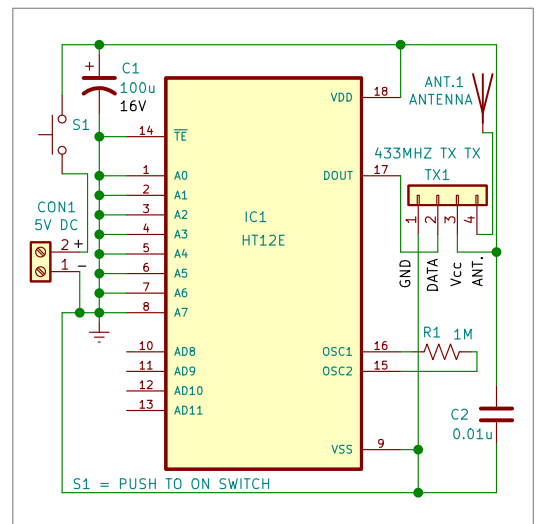


Fig. 3: Circuit diagram of the transmitting unit